

School Bell Automation System using FPGA

Problem Statement:

Design and implement a School Bell Automation System on an FPGA board that automatically turns ON an LED to represent the bell at specific times of the day, based on a predefined schedule. The system should simulate a real-time digital clock, ring the bell using an LED for each scheduled event, and optionally support manual bell activation.

Objectives:

1. Simulate a real-time digital clock on FPGA.
2. Pre-program a bell schedule with configurable time slots.
3. Turn ON an LED (representing a bell) when the current time matches any bell time.
4. Optional: Provide manual LED bell activation using a push button.
5. Use FSM-based logic for time tracking and bell control.

Inputs:

Input	Description
clk	System clock (used to derive 1Hz real-time clock)
reset	Synchronous reset for the whole system
enable	To start the system
manual_ring_btn	Button to manually trigger the bell (LED)

Outputs :

Output	Description
bell_led	LED to represent the ringing bell
curr_time	7-segment display (or LEDs) to show current time

Bell schedule:

Time (HH:MM)	Bell Event
09:00	Start of school
09:45	End of Period 1
10:30	End of Period 2
11:15	Break Time
11:30	End of Break
12:15	End of Period 3
01:00	End of Period 4
01:45	Lunch Break
02:15	End of Lunch Break
03:00	School Ends